

$$e^{-\frac{i}{\hbar} \hat{H} t} |\psi\rangle$$

$$e^{-\frac{i}{\hbar} \hat{H}_0 t} |\phi_{in}\rangle$$

$$0$$

$$|\psi\rangle$$

$$e^{-\frac{i}{\hbar} \hat{H} t} |\psi\rangle$$

$$e^{-\frac{i}{\hbar} \hat{H}_0 t} |\phi_{out}\rangle$$

$$e^{-\frac{i}{\hbar} \hat{H}_0 t} |\phi_{in}\rangle = \int_{-\infty}^{+\infty} dp \Phi_{in}(p) |\psi_p^{FREE}\rangle e^{-\frac{i}{\hbar} \frac{p^2}{2m} t}$$

$$|\psi(t)\rangle = e^{-\frac{i}{\hbar} \hat{H} t} |\psi\rangle$$

$$|\psi\rangle = \int_{-\infty}^{+\infty} dp \Phi_{in}(p) |\psi_p^+\rangle$$

$$\langle x | \psi_p^{FREE} \rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{+\frac{i}{\hbar} p x} = \psi_p^{FREE}(x)$$

$$\hat{V} = V(\hat{x})$$

$$\underline{M} \vec{c}_j = m_j \vec{c}_j \quad j=1,2,3,\dots,10$$

$$f(\underline{M}) = \sum_j \vec{c}_j f(m_j) \vec{c}_j^{*T}$$

$$|\psi_p^+\rangle = |\psi_p^{FREE}\rangle + \frac{1}{E - \hat{H}_0 + i\varepsilon} \hat{V} |\psi_p^{FREE}\rangle$$

RETARDED GREEN OPERATOR

LIPPMANN-SCHWINGER EQUATION
(EXPLICIT FORM)

$$E = \frac{p^2}{2m}$$

$$\hat{G}_E^+(E - \hat{H}_0 + i\varepsilon) = \hat{1}$$

$\varepsilon \rightarrow 0_+$

$G_E^+(x, x')$ RETARDED GREEN FUNCTION

$$\psi_p^+(x) = \psi_p^{FREE}(x) + \int_{-\infty}^{+\infty} dx' \langle x | \frac{1}{E - \hat{H}_0 + i\varepsilon} | x' \rangle V(x') \psi_p^{FREE}(x')$$

$$(E - \hat{H}_0 + i\varepsilon) |\psi_p^+\rangle = (E - \hat{H}_0 + i\varepsilon) |\psi_p^{FREE}\rangle + \hat{V} |\psi_p^{FREE}\rangle$$

TWO CONSEQUENCES:

$$\textcircled{1} (E - \hat{H}_0 + i\varepsilon) |\psi_p^+\rangle = (E - \hat{H}_0 + i\varepsilon) |\psi_p^{FREE}\rangle = |\phi\rangle$$

$$\hat{H} |\psi_p^+\rangle = E |\psi_p^+\rangle$$

$$E = \frac{p^2}{2m}$$

$$\hat{H}_0 |\psi_p^{FREE}\rangle = E |\psi_p^{FREE}\rangle$$

$$\textcircled{2} (E - \hat{H}_0 + i\varepsilon) |\psi_p^+\rangle = (E - \hat{H}_0 + i\varepsilon) |\psi_p^{FREE}\rangle + \frac{1}{E - \hat{H}_0 + i\varepsilon} \hat{V} |\psi_p^+\rangle$$

$$G_{E_0}^\pm(x, x') =$$

$$= \langle x | \frac{1}{E - \hat{H}_0 + i\varepsilon} | x' \rangle$$

$$f(\hat{H}_0) = \int_{-\infty}^{+\infty} dp |\psi_p^{FREE}\rangle \left(\frac{p^2}{2m} \right) \langle \psi_p^{FREE} |$$

$$\frac{1}{E - \hat{H}_0 + i\varepsilon} = \int_{-\infty}^{+\infty} dp' \frac{|\psi_{p'}^{FREE}\rangle \langle \psi_{p'}^{FREE}|}{E - \frac{p'^2}{2m} + i\varepsilon}$$

$$G_{E_0}^+(x, x') = \int_{-\infty}^{+\infty} dp' \frac{\psi_p^{FREE}(x) \psi_{p'}^{FREE*}(x')}{E - \frac{p'^2}{2m} + i\varepsilon}$$

$$= \frac{1}{2\pi\hbar} \int_{-\infty}^{+\infty} dp' \frac{e^{+\frac{i}{\hbar} p'(x-x')}}{E - \frac{p'^2}{2m} + i\varepsilon} =$$

$$= \dots \dots \dots$$

$$G_{E_0}^\pm(x, x') = \mp i \frac{m}{\hbar^2 k} e^{\pm i k |x-x'|}$$

$$\hbar k = \sqrt{2mE}$$

$$e^{-\frac{i}{\hbar} \hat{H} t} |\psi\rangle$$

$$e^{-\frac{i}{\hbar} \hat{H}_0 t} |\phi_{in}\rangle$$

$$0$$

$$|\psi\rangle$$

$$e^{-\frac{i}{\hbar} \hat{H} t} |\psi\rangle$$

$$e^{-\frac{i}{\hbar} \hat{H}_0 t} |\phi_{out}\rangle$$

$$e^{-\frac{i}{\hbar} \hat{H}_0 t} |\phi_{in}\rangle = \int_{-\infty}^{+\infty} dp \Phi_{in}(p) |\psi_p^{FREE}\rangle e^{-\frac{i}{\hbar} \frac{p^2}{2m} t}$$

$$|\psi(t)\rangle = e^{-\frac{i}{\hbar} \hat{H} t} |\psi\rangle$$

$$\langle x | \psi_p^{FREE} \rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{+\frac{i}{\hbar} p x} = \psi_p^{FREE}(x)$$

$$\hat{V} = V(\hat{x})$$

$$|\psi_p^+\rangle \doteq |\psi_p^{FREE}\rangle \quad \text{BORN SERIES}$$

$$|\psi_p^+\rangle \doteq |\psi_p^{FREE}\rangle + \frac{1}{E - \hat{H}_0 + i\epsilon} \hat{V} |\psi_p^{FREE}\rangle$$

$$+ \frac{1}{E - \hat{H}_0 + i\epsilon} \hat{V} \frac{1}{E - \hat{H}_0 + i\epsilon} \hat{V} |\psi_p^{FREE}\rangle$$

$$+ \dots$$

$$|\psi_p^+\rangle = |\psi_p^{FREE}\rangle + \frac{1}{E - \hat{H}_0 + i\epsilon} \hat{V} |\psi_p^+\rangle \quad \text{LIPPMANN-SCHWINGER EQUATION (IMPLICIT FORM)}$$

$$? \psi_p^+(x \rightarrow \pm\infty)$$

$$p > 0$$

$$\psi_p^+(x) = \psi_p^{FREE}(x) + \int_{-\infty}^{+\infty} dy \langle x | \frac{1}{E - \hat{H}_0 + i\epsilon} | y \rangle V(y) \psi_p^+(y)$$

$$\psi_p^+(x) = \frac{1}{\sqrt{2\pi\hbar}} e^{+\frac{i}{\hbar} p x} + \int_a^b dy (-i) \frac{m}{\hbar p} e^{+\frac{i}{\hbar} p |x-y|} V(y) \psi_p^+(y)$$

VALID FOR ANY $x \in \mathbb{R}$

$$x \rightarrow +\infty$$

$$\psi_p^+(x) = \frac{1}{\sqrt{2\pi\hbar}} e^{+\frac{i}{\hbar} p x} + \int_a^b dy \frac{(-i)m}{\hbar p} \frac{1}{\sqrt{2\pi\hbar}} e^{+\frac{i}{\hbar} p (x-y)} V(y) \psi_p^+(y)$$

$$= \frac{1}{\sqrt{2\pi\hbar}} e^{+\frac{i}{\hbar} p x}$$

$$1 - \frac{i m}{\hbar p} \sqrt{2\pi\hbar} \int_a^b dy e^{-\frac{i}{\hbar} p y} V(y) \psi_p^+(y)$$

$$\psi_p^+(x \rightarrow \infty) = \frac{1}{\sqrt{2\pi\hbar}} e^{+\frac{i}{\hbar} p x} T(p) \quad T(p)$$

$$G_{E_0}^{\pm}(x, x') =$$

$$= \langle x | \frac{1}{E - \hat{H}_0 + i\epsilon} | x' \rangle$$

$$f(\hat{H}_0) = \int_{-\infty}^{+\infty} dp |\psi_p^{FREE}\rangle \left(\frac{p^2}{2m} \right) \langle \psi_p^{FREE} |$$

$$\frac{1}{E - \hat{H}_0 + i\epsilon} = \int_{-\infty}^{+\infty} dp' \frac{|\psi_{p'}^{FREE}\rangle \langle \psi_{p'}^{FREE}|}{E - \frac{p'^2}{2m} + i\epsilon}$$

$$G_{E_0}^+(x, x') = \int_{-\infty}^{+\infty} dp' \frac{\psi_p^{FREE}(x) \psi_{p'}^{FREE*}(x')}{E - \frac{p'^2}{2m} + i\epsilon}$$

$$= \frac{1}{2\pi\hbar} \int_{-\infty}^{+\infty} dp' \frac{e^{+\frac{i}{\hbar} p'(x-x')}}{E - \frac{p'^2}{2m} + i\epsilon} =$$

$$= \dots$$

$$G_{E_0}^{\pm}(x, x') = \mp i \frac{m}{\hbar k} e^{\pm i k |x-x'|}$$

$$\hbar k = \sqrt{2mE} = p$$

$$2. \psi_p^+(x \rightarrow \pm\infty)$$

$$p > 0$$

$$\psi_p^+(x) = \psi_p^{\text{FREE}}(x) + \int_{-\infty}^{+\infty} dy \langle x | \frac{1}{E - \hat{H}_0 + i\varepsilon} | y \rangle V(y) \psi_p^+(y)$$

$$\psi_p^+(x) = \frac{1}{\sqrt{2\pi k}} e^{+i k x} + \int_a^b dy \frac{(-i)m}{k p} e^{+i k |x-y|} V(y) \psi_p^+(y)$$

VALID FOR ANY $x \in \mathbb{R}$

$$x \rightarrow -\infty$$

$$\psi_p^+(x) = \frac{1}{\sqrt{2\pi k}} e^{+i k x} + \int_a^b dy \frac{(-i)m}{k p} e^{+i k |x-y|} V(y) \psi_p^+(y)$$

$$= \frac{1}{\sqrt{2\pi k}} e^{+i k x} + \frac{1}{\sqrt{2\pi k}} e^{+i k x} R(p)$$

$$R(p) = \left(\frac{1}{\sqrt{2\pi k}} \int_a^b dy \frac{(-i)m}{k p} e^{+i k |x-y|} V(y) \psi_p^+(y) \right)$$

$$G_{E_0}^{\pm}(x, x') = \langle x | \frac{1}{E - \hat{H}_0 + i\varepsilon} | x' \rangle$$

$$f(\hat{H}_0) = \int_{-\infty}^{+\infty} dp \left| \psi_p^{\text{FREE}} \right\rangle \left\langle \psi_p^{\text{FREE}} \right|$$

$$\frac{1}{E - \hat{H}_0 + i\varepsilon} = \int_{-\infty}^{+\infty} dp \frac{|\psi_p^{\text{FREE}}\rangle \langle \psi_p^{\text{FREE}}|}{E - \frac{p^2}{2m} + i\varepsilon}$$

$$G_{E_0}^{\pm}(x, x') = \int_{-\infty}^{+\infty} dp \frac{\psi_p^{\text{FREE}}(x) \psi_p^{\text{FREE}*}(x')}{E - \frac{p^2}{2m} + i\varepsilon}$$

$$= \frac{1}{2\pi k} \int_{-\infty}^{+\infty} dp e^{+i k |x-x'|} \frac{1}{E - \frac{p^2}{2m} + i\varepsilon} = \dots$$

$$G_{E_0}^{\pm}(x, x') = \mp i \frac{m}{k} e^{+i k |x-x'|}$$

$$k = \sqrt{2mE} = p$$

$$\hat{H} = \hat{H}^\dagger$$

$$\langle \psi(1) | \hat{A} \psi(2) \rangle = \langle \hat{A}^\dagger \psi(1) | \psi(2) \rangle$$

HERMITIAN MATRIX

$$\underline{\underline{H}} = \underline{\underline{H}}^\dagger$$

$$\begin{pmatrix} 1 & 3+7i \\ 3-7i & 2 \end{pmatrix}$$

$$\langle \psi(1) | \psi(2) \rangle = \int_{-\infty}^{+\infty} dx \psi(1)^*(x) \psi(2)(x)$$

NONHERMITIAN

$$\lambda \in \mathbb{C} \quad (\psi(1) | \psi(2)) =$$

$$= \int_{-\infty}^{+\infty} dx \psi(1)(x) \psi(2)(x)$$

$$\underline{\underline{H}} = \begin{pmatrix} +1 & i \\ i & -1 \end{pmatrix}$$

$$\underline{\underline{H}} \vec{c} = \varepsilon \vec{c}$$

EXCEPTIONAL
POINTS

